



# Azure Community Conference 2021

*India's largest Azure Conference*



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# Dennie Declercq

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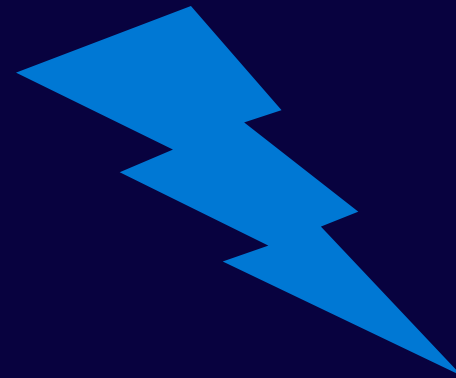
President, Developer, MVP @ DDSoft

#AzConfDev



# AI for The Better (Life of People with Disabilities)

Lightning Edition

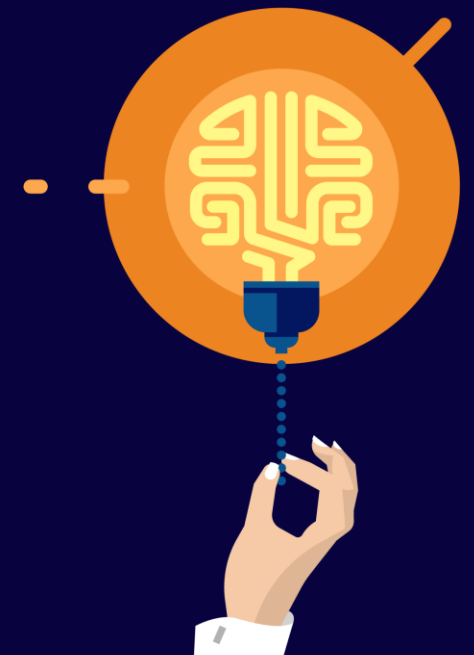


# Content



- Primer into Accessibility
- Microsoft Cognitive Services API
- Problems that People With Disabilities can have
- Demo's, Demo's, Demo's!

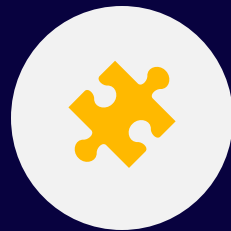
# Primer into Accessibility



# Kind of disabilities



Learning  
disability



Autism



Visual



Auditory



Motoric



Much more...

# Learning disability

Significant limitations in both:

- Intellectual functioning
- Adaptive behavior



# Learning Disability

## Intellectual Functioning

Also called intelligence

Mental capacity

IQ score between 70-75

## Adaptive Behavior

### Conceptual Skills

- Language/literacy
- Money
- Time/numbers

### Social Skills

- Social responsibility
- Self-esteem

### Practical skills

- Activities daily living
- Personal care / healthcare

# Principles of Inclusion



- Same activities
- Same district
- Work Together
- Adjustments par person

# Digital Difficulties for PWD.

## Learning Disability

- Read
- Understand

## Autism

- Understand
- See structure

## Blindness or vision loss

- Decreased ability to read
- Decreased ability to navigate

## Auditory disabilities

- Hearing notifications
- Following movies without audio description.





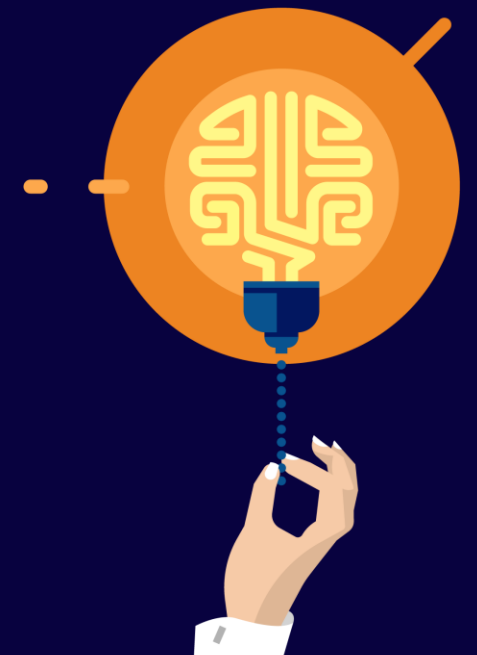
# Our Focus

People with Learning  
disabilities

Autism

People with mental health  
situations

# Microsoft Cognitive Services API's



# What is Microsoft Cognitive Services?



Part of Microsoft's AI platform:

No need-to-know AI

No need-to-know deep math algorithms

Just perform easy AI calls

Possibilities: REST style/ SDK Style\*

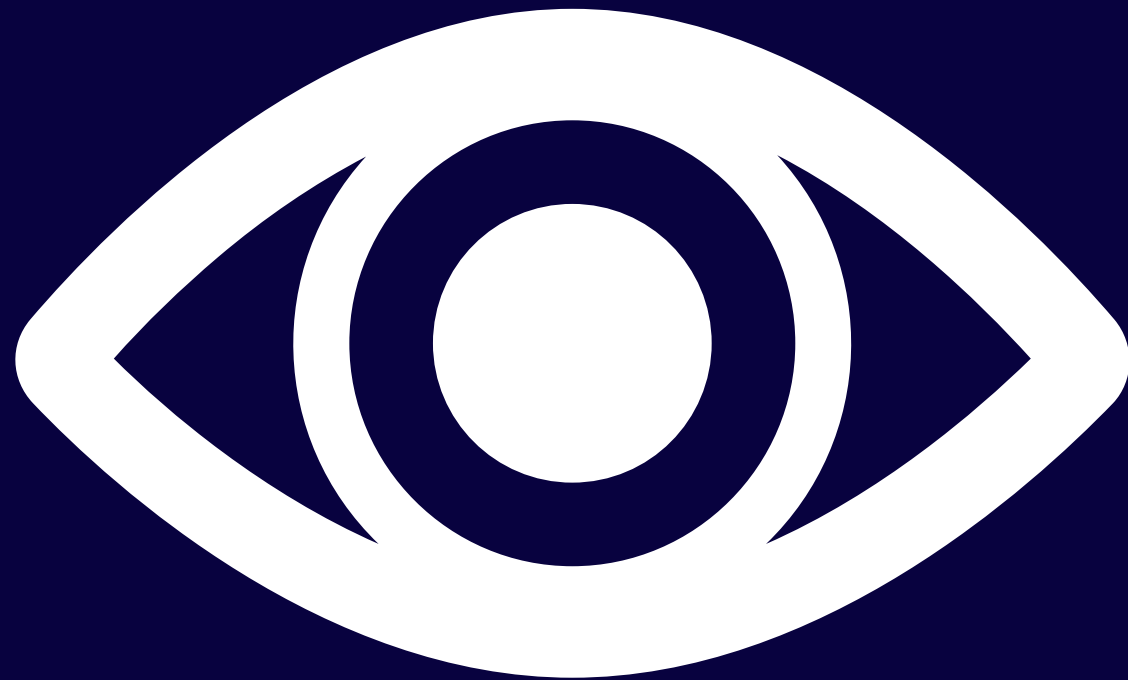
\*Not for each Cognitive Service (yet)



# Speech Service

Text-to-speech

Custom Voice

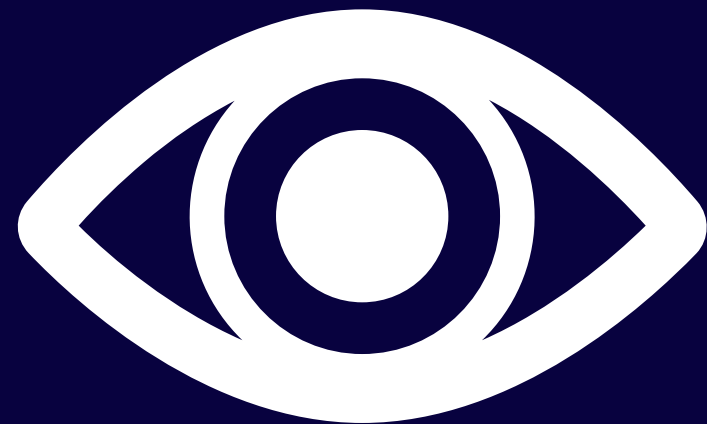


# Vision API

Analyzes images

Let PWD take pictures and tells what it is.





# Face API

Detect Emotion

Detect Facial Hair

Detect Smile

Detect Age

# BOT Service

Provide BOTs

BOTs really can assist PWD

Make clear guidelines





# Immersive Reader

Accessibility features like:

- Reading aloud websites

- Linking pictographs to websites

- Text appearance options



# Translator Service

Translation

Language Detection

# Problems that People With Disabilities can have



# Literacy

Reading information  
Understanding written text  
Recognizing money  
Math





# Executive Functioning

Planning 'simple' tasks  
Making choices  
Concentration



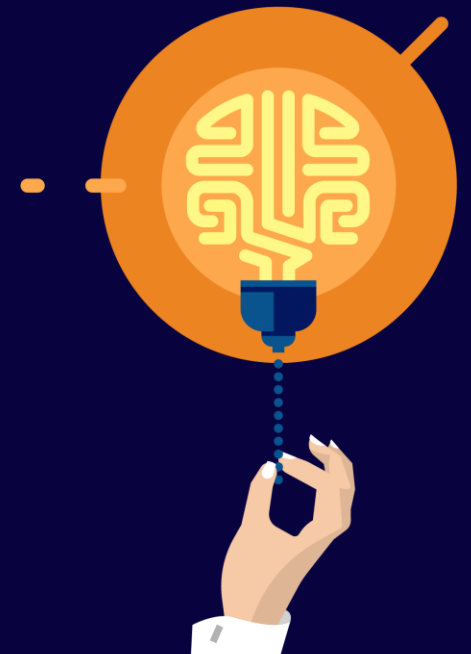
# Trust & Self-Trust

Self-trust can be hard  
Can I really do this?  
Will this really work?



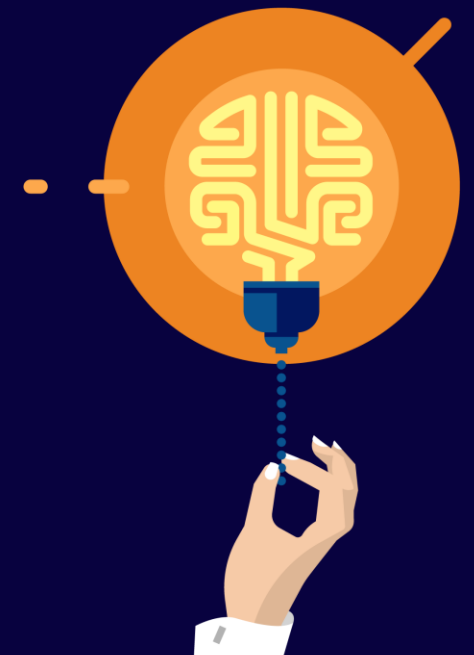


# Demo's



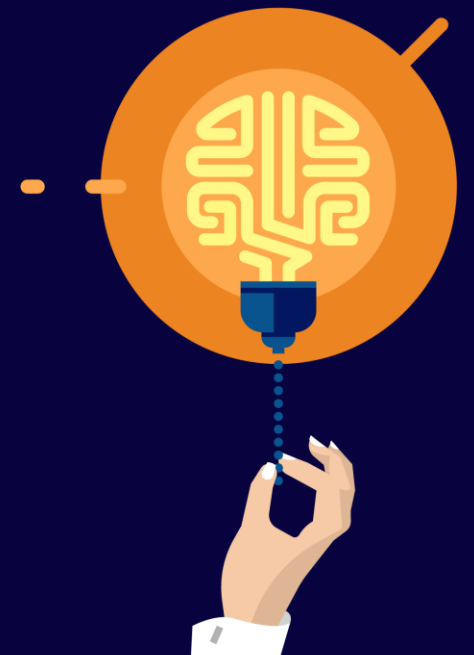
# DEMO: Creating a Cognitive Speech Service

<https://go.ddsoft.be/texthelp>



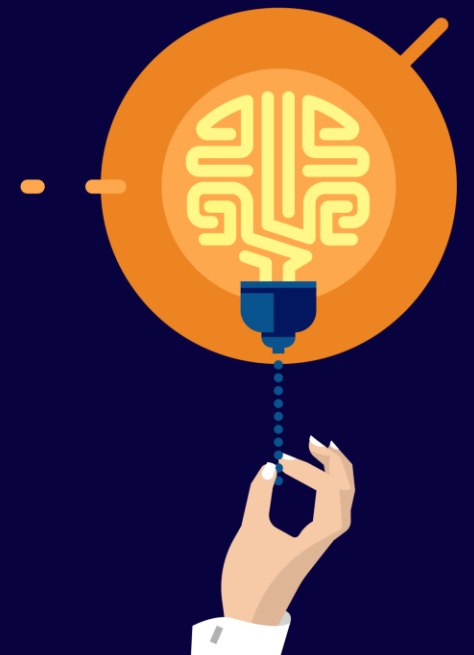
# DEMO: Speech Service in Action

<https://go.ddsoft.be/texthelp>

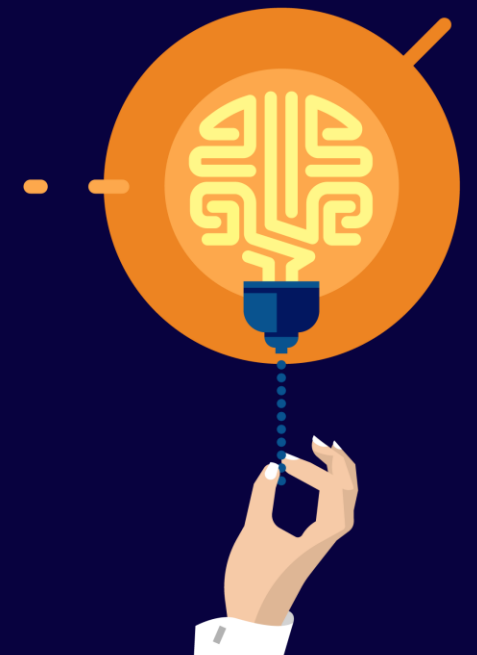


# DEMO: Translate Service in Action

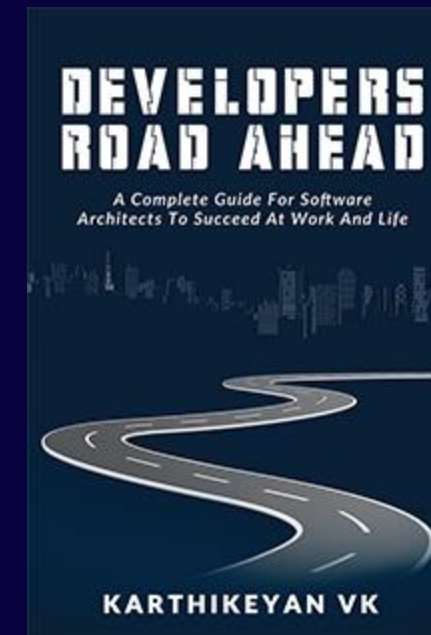
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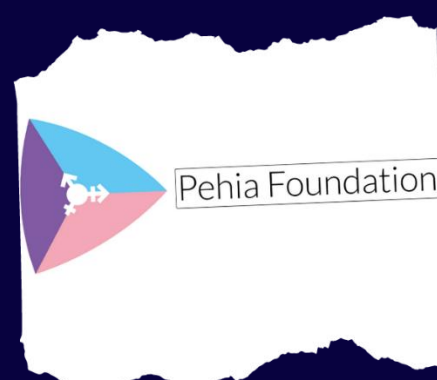
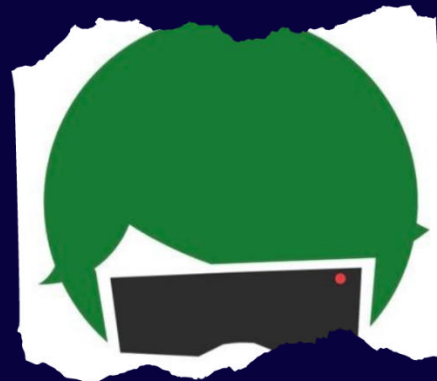
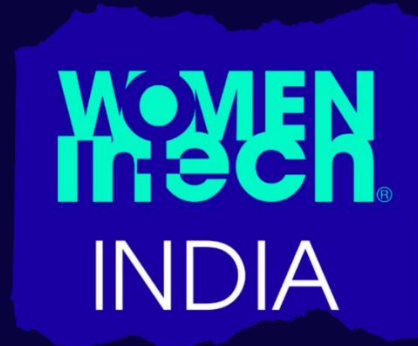
# DEMO: The Future of Web Accessibility



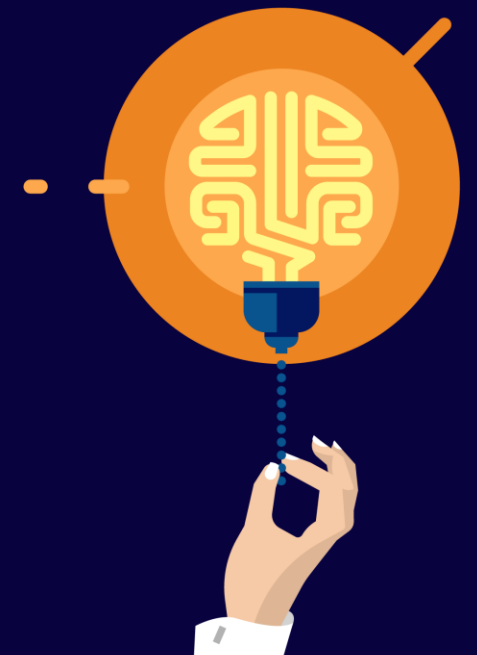
# Our Partners



# Communities

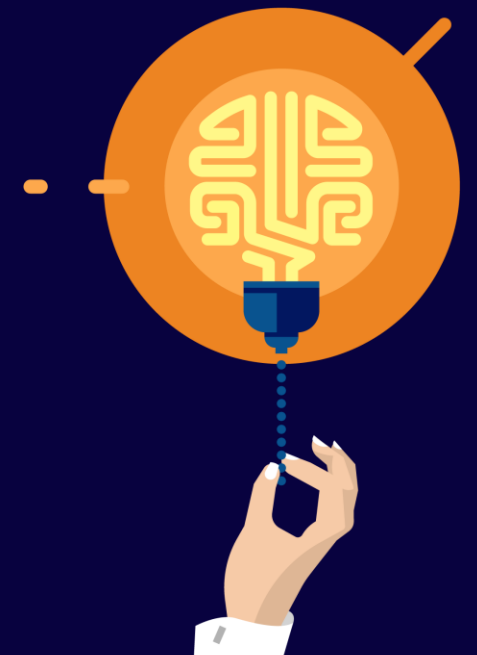


# Q & A





# Feedback



# THANK YOU!



**Dennie Declercq**  
@DennieDeclercq  
#DreamingIsBelieving



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